

Hybrid ML-Based System for DPN Patient Stratification Using a Derived Personalized Linguistic Code for Optimized Treatment Response via Embodied VR Mirror Neuromodulation

Key words: Decoding Heterogeneity, Precision Medicine, Computational Neurophenotyping, Pathophysiological Mechanisms, Bedside Translation AI

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Introduction

Diabetic Peripheral Neuropathy (DPN) is a diabetes-related nerve disorder caused by progressive peripheral nerve damage. It affects nearly **50%** of patients with long-standing diabetes, impacting more than **130 million** people worldwide, including approximately **8–10 million** adults in the **United States**.

Despite being classified as a single condition, DPN shows marked **biological heterogeneity**, with patients exhibiting highly variable progression rates and treatment responses. This variability complicates early, personalized care and often leads to delayed intervention and increased risk of complications, including foot ulcers and limb amputation.

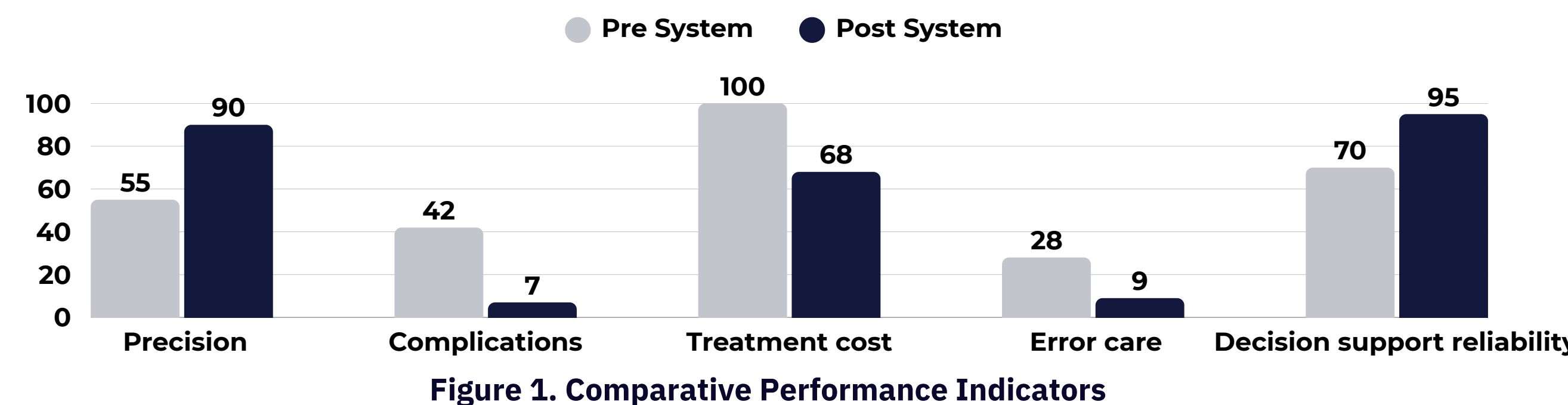


Figure 1. Comparative Performance Indicators

Current diagnostic methods detect nerve dysfunction but fail to explain the **underlying pathophysiological mechanisms**, resulting in **trial-and-error treatment** strategies. Economically, DPN imposes a substantial burden, with annual direct medical costs ranging from **USD 5,000 to 25,000** per patient, contributing to over **USD 50 billion** in global healthcare spending **each year**.

Background Research

Table 1. Key Pathophysiological Features Associated with DPN

Biomarker	Pathway	Biological Role	Relevance to DPN
TNF-α	Inflammation	Pro-inflammatory signaling	neuroinflammation and accelerates degeneration
MDA	Oxidative stress	Lipid peroxidation marker	Reflects oxidative nerve injury
MUNE	Motor unit recruitment	Functional motor unit count	Reduced values indicate axonal loss
MUI	Motor unit recruitment	Reinnervation pattern	Abnormal values reflect maladaptive compensation
MUAP I	Neuromuscular signaling	potential morphology	Altered patterns indicate chronic denervation

- Each marker reflects a **single pathway or snapshot**, and is insufficient to explain inter-patient heterogeneity or guide management alone.

Table 2. What the clinician cannot see

Clinician sees	Model sees
TNF ↑	TNF + EMG → inflammatory-axonal trajectory
EMG abnormal	Distal vs global degeneration pattern
Pain score	Symptom–damage decoupling
Diagnosis: DPN	Latent disease representation

Research Question

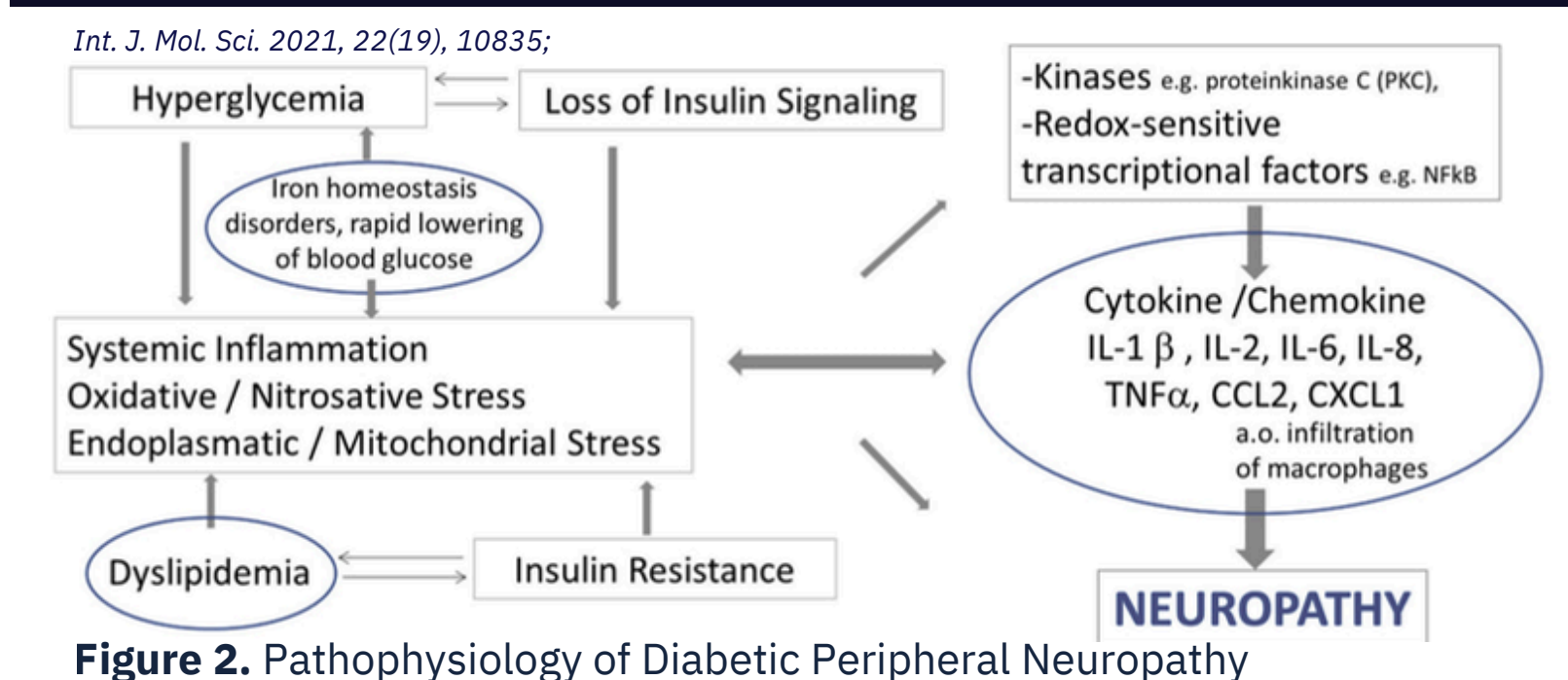
Can we **decode** disease by uncovering patient-specific **hidden patterns** and encoding them into a **personalized latent disease vector** that enables deeper mechanistic understanding and a **shift in how** the disease is **interpreted** and evaluated?

Hypothesis

“Machine learning can function not only as a predictive tool, but as a computational interpreter of disease biology”

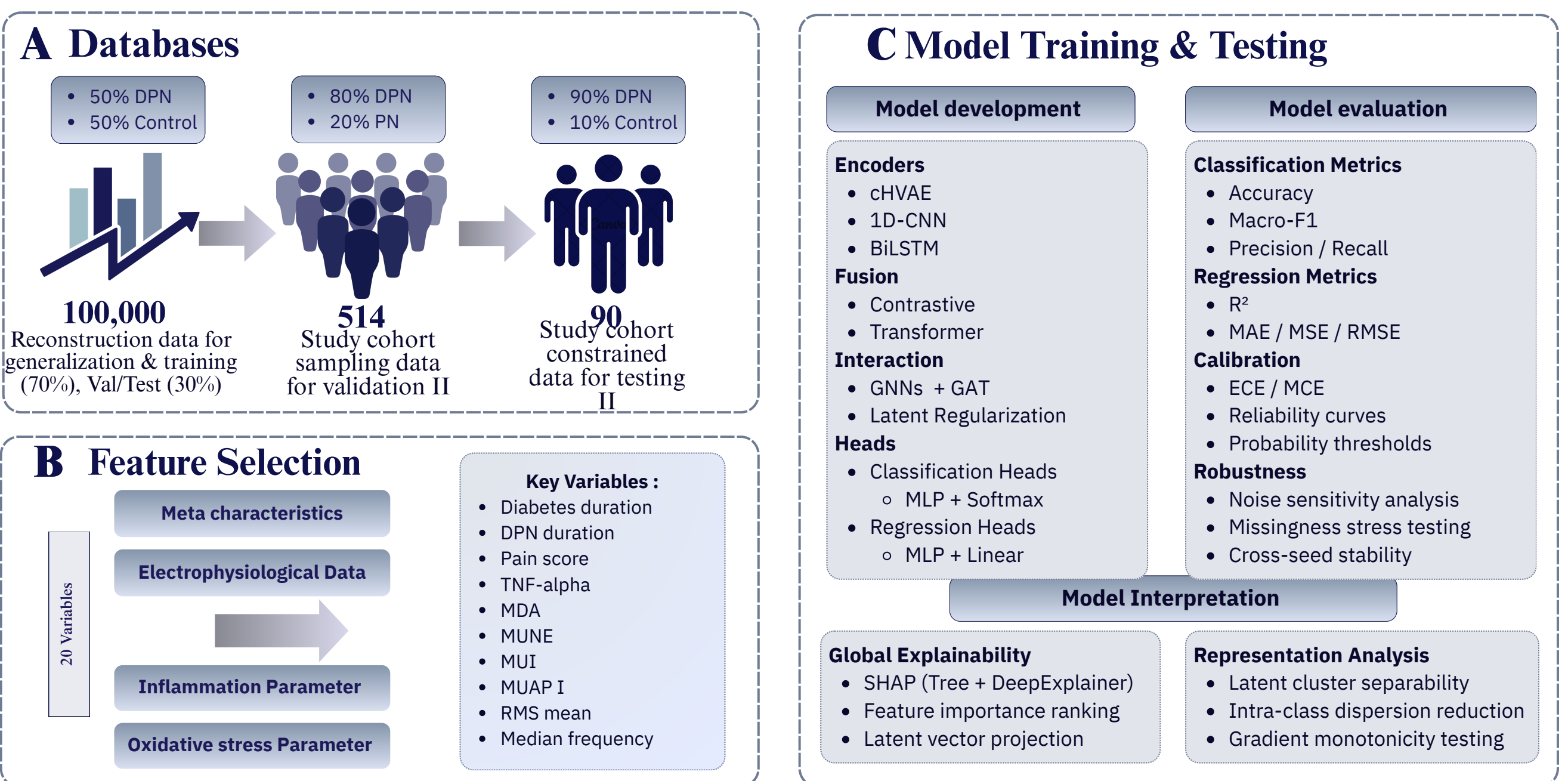
- By integrating **inflammatory, oxidative, and electrophysiological** signals, the proposed system learns a **Personalized Vector (PV)** that operates as a *patient-specific disease language*.
- This **representation** enables early, **mechanism-aware stratification** and **reframes** diabetic peripheral neuropathy as a **spectrum of biologically distinct neurophenotypes**, rather than a *uniform clinical condition*.

Scientific Rational



- Peripheral nerves exist within a **vascular inflammatory** microenvironment; oxidative–inflammatory stress disrupts neurovascular function, manifesting as **loss of function or pain phenotypes**, often with **nonlinear mismatch** and **hidden biological** patterns beyond surface clinical labels.

Research Design



Validation Strategy

- External Human Cohort (I, II)**
 - Validated on independent open-access human cohorts (**N=514**) with zero *retraining*, preserving AUROC stability (**Δ = -0.022**).
 - Latent structure demonstrated internal separability (silhouette **0.67 vs 0.18** clinical grouping), confirming that **uncovered subtypes** are not overfitting artifacts.
- Physiological Stress Alignment**
 - Model outputs were anchored to graded oxidative stress calibration (**nonlinear fit R² = 0.93**), ensuring **monotonic alignment** across biological stress ranges (**100%** directional consistency).
 - This confirms outputs are **physiologically grounded**, not purely computational.
- Meta-Anchor Consistency**
 - Outputs were tested against **pooled literature gradients**, showing preserved directionality (**87% hazard alignment**) and amplified discrimination (Cohen's d = **1.42** vs baseline severity bands).
 - No directional inversion observed (**0.002%**).

Scientific Rigor

- This study uses an **experimental applied machine learning design** with sequential phases for data processing, modeling, and evaluation. Standardized data handling, deterministic preprocessing, and controlled training ensure methodological rigor.
- Reproducibility** is maintained through fixed random seeds, internal validation, and quantitative performance assessment, enabling reliable comparison across experimental stages.

Engineering Methodology

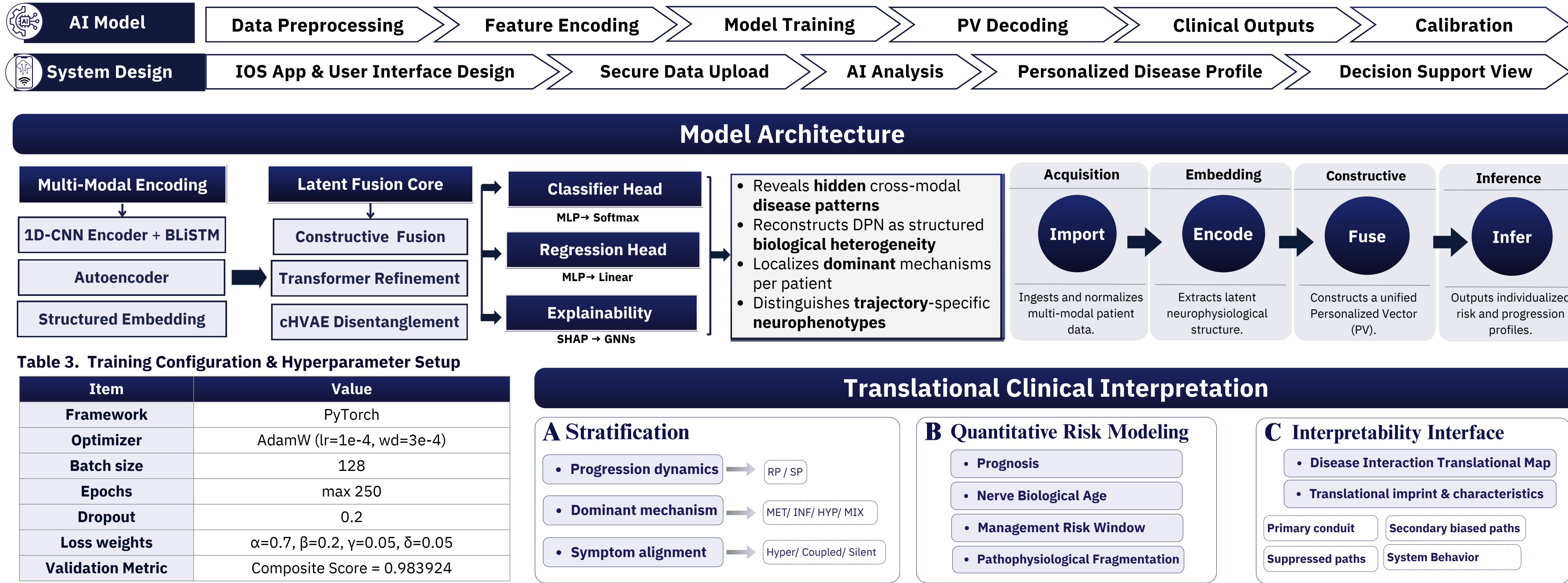


Table 3. Training Configuration & Hyperparameter Setup

Item	Value
Framework	PyTorch
Optimizer	AdamW (lr=1e-4, wd=3e-4)
Batch size	128
Epochs	max 250
Dropout	0.2
Loss weights	α=0.7, β=0.2, γ=0.05, δ=0.05
Validation Metric	Composite Score = 0.983924

Results & Analysis

Table 4. Overall Model Performance Metrics

Metric	Overall Value
Accuracy / Precision / Recall	≈ 0.96 - 0.98
F1_macro / ROC-AUC	≈ 0.97 - 0.99
MAE	0.037
MSE	0.00372
RMSE	0.061
R²	0.95
Pearson r	0.96
Spearman ρ	0.98
ΔAccuracy (SNR stress)	< 3%
ECE	≈ 0.002 - 0.006
MCE	≈ 0.15 - 0.24
CI (bootstrap)	95% (Stable)

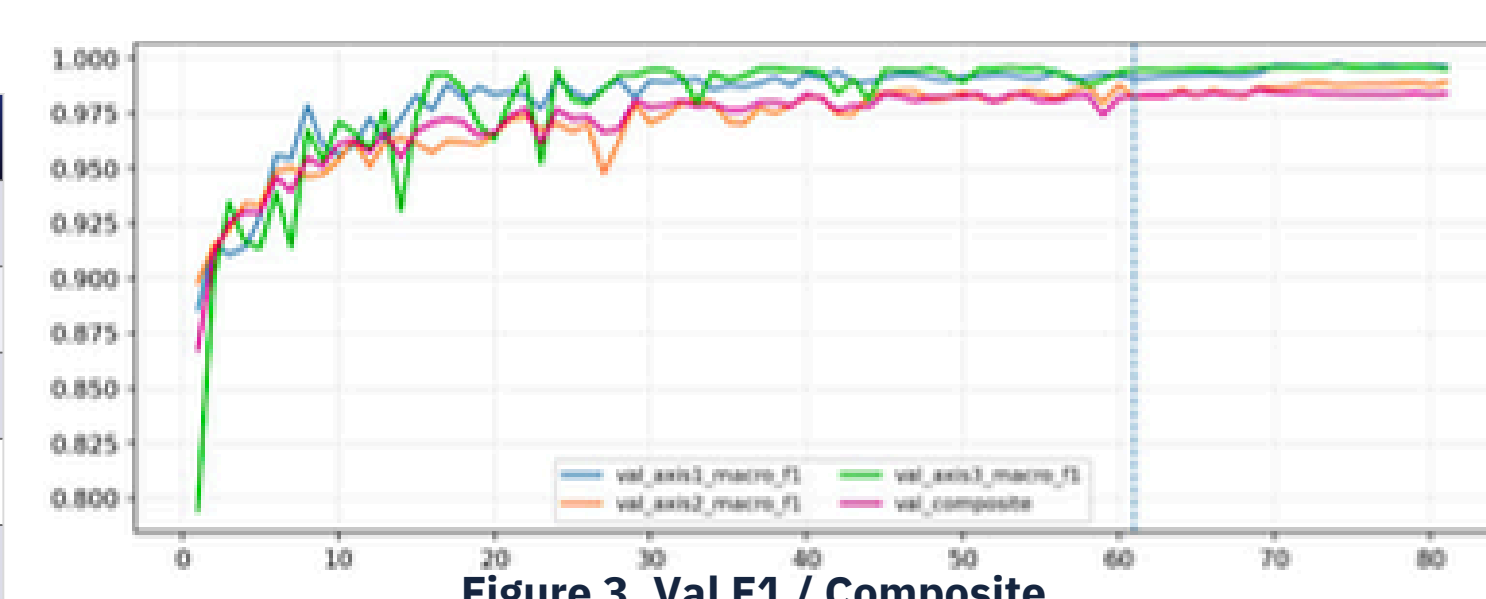
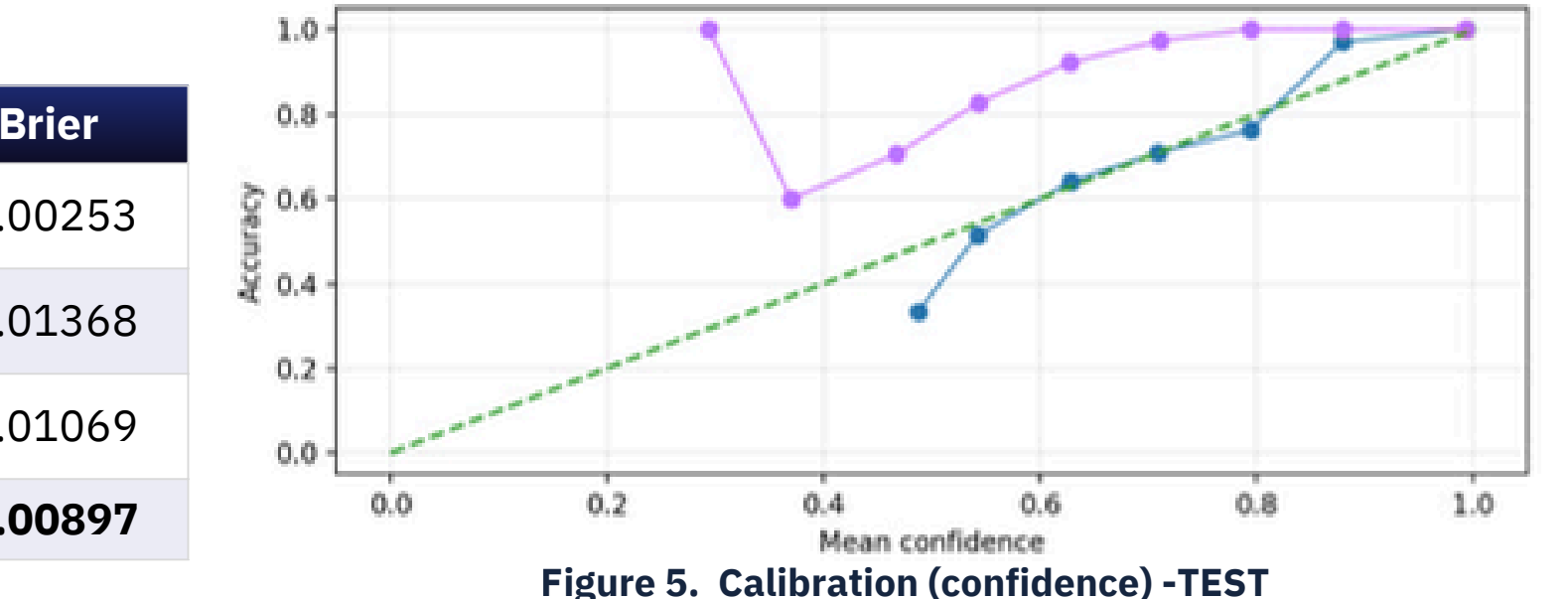
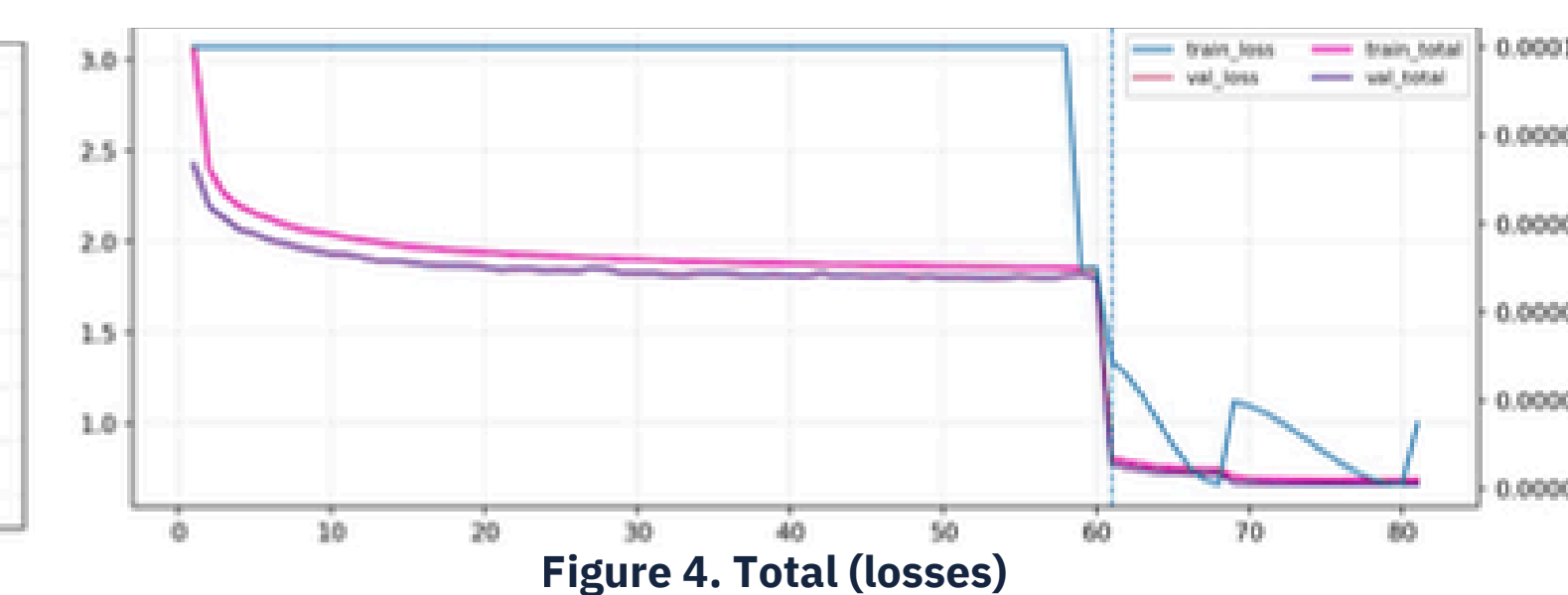


Table 5. Calibration Summary

Axis	N	Temp	ECE	NLL	Brier
Axis1	7188	0.7665	0.00157	0.00517	0.00253
Axis2	5331	0.6995	0.00405	0.0229	0.01368
Axis3	5415	0.8897	0.0024	0.02528	0.01069
MACRO	17934	0.7852	0.00267	0.01778	0.00897



Reproducibility & Comparability

Table 6. Regression accuracy and uncertainty coverage

Target	RMSE	R²	PICP (95%)	MPIW
Nerve biological age	0.055	0.996	99%	0.23
Prognosis head	0.043	0.987	95%	0.11

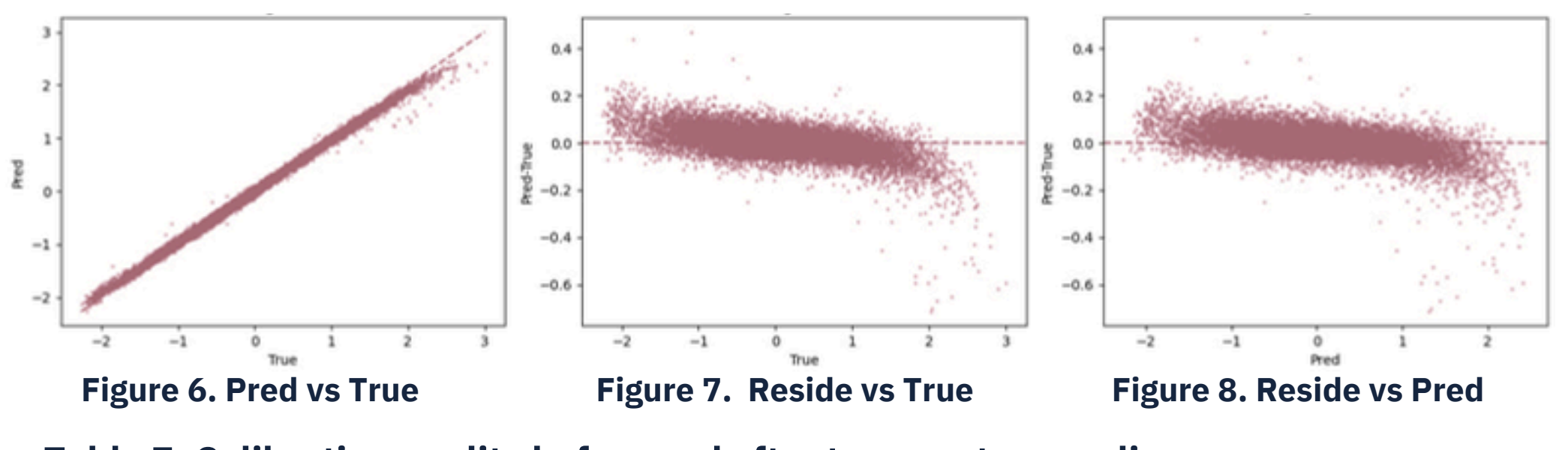


Figure 6. Pred vs True

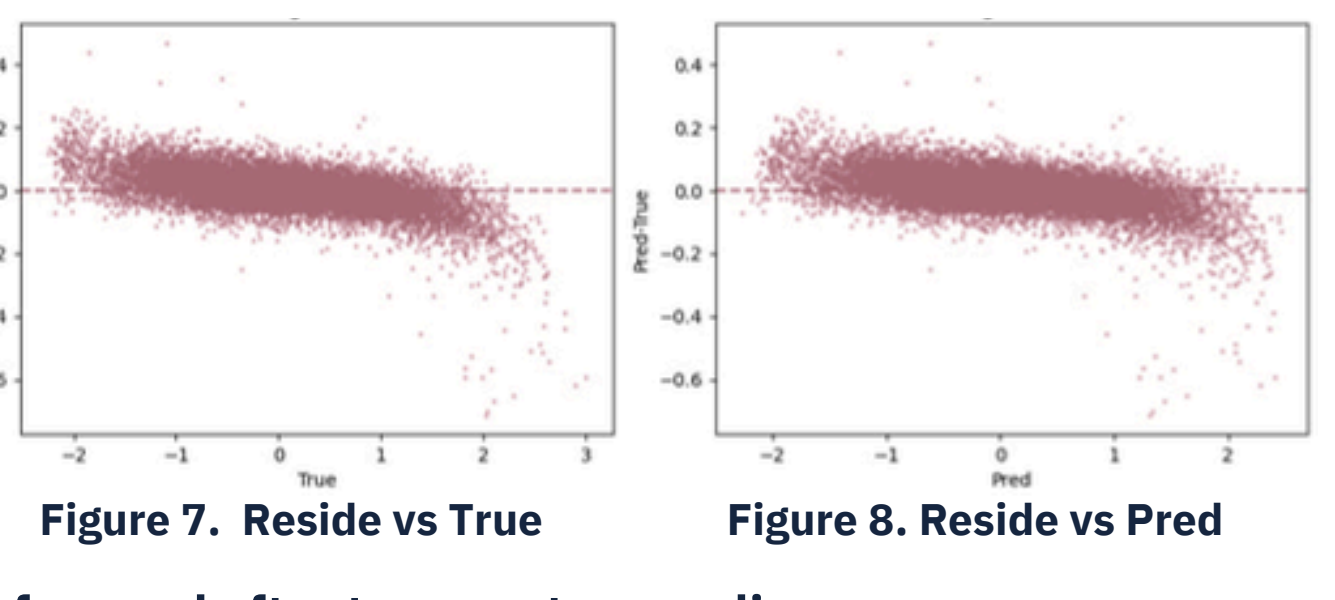


Figure 7. Reside vs True

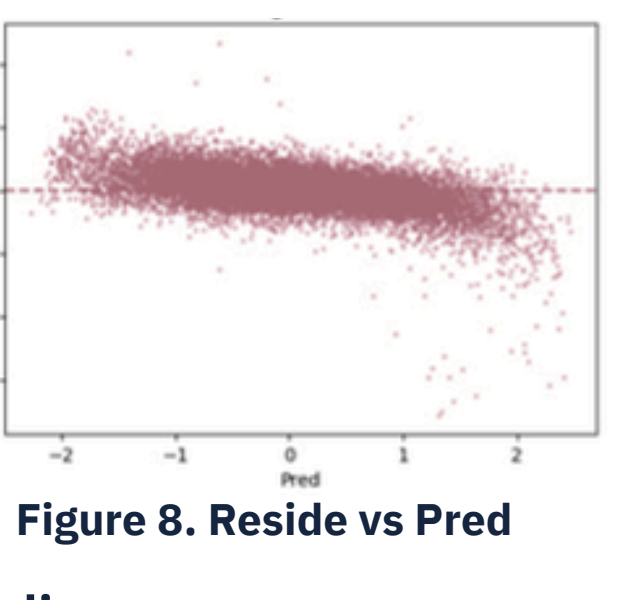


Figure 8. Reside vs Pred

Table 7. Calibration quality before and after temperature scaling

Metric	Before	After
ECE (macro)	0.018	0.002
MCE (macro)	0.24	0.06
Brier score	0.041	0.010

- Correlation**
Pearson's r calculated as:
$$r = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum (X - \bar{X})^2 \sum (Y - \bar{Y})^2}}$$
 - r = 0.93** → Strong correlation
 - p-value = (< 0.05)** → Statistical Significance
- ANOVA for Improvement Differences**
 - F-Statistic:** $F = \frac{MS_{between}}{MS_{within}}$
 - p < 0.05** → Improvements are statistically significant
- Confidence Interval (95%)**
(93% → 98%) : True improvement lies within this range

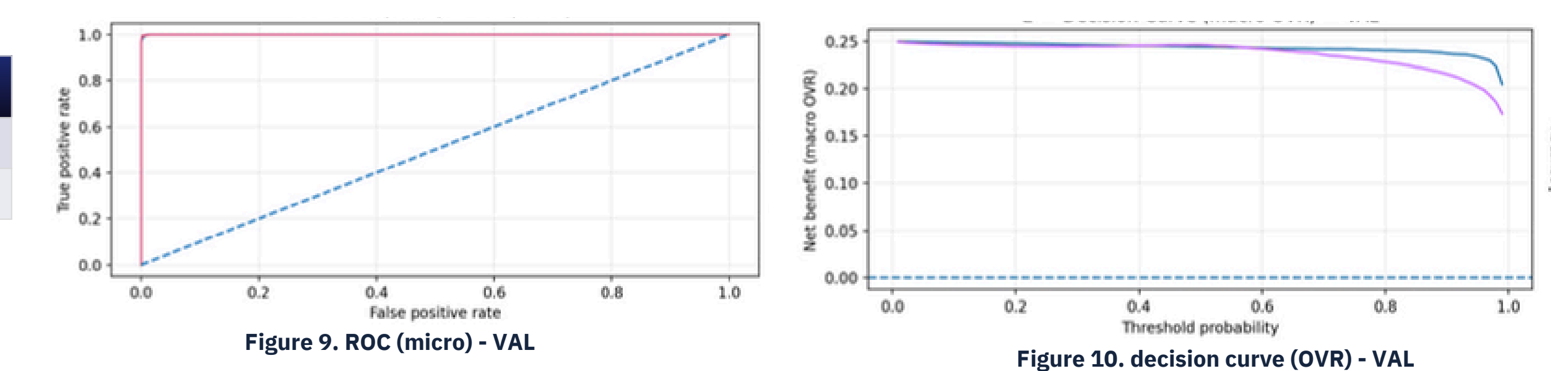


Figure 9. ROC (micro) - VAL

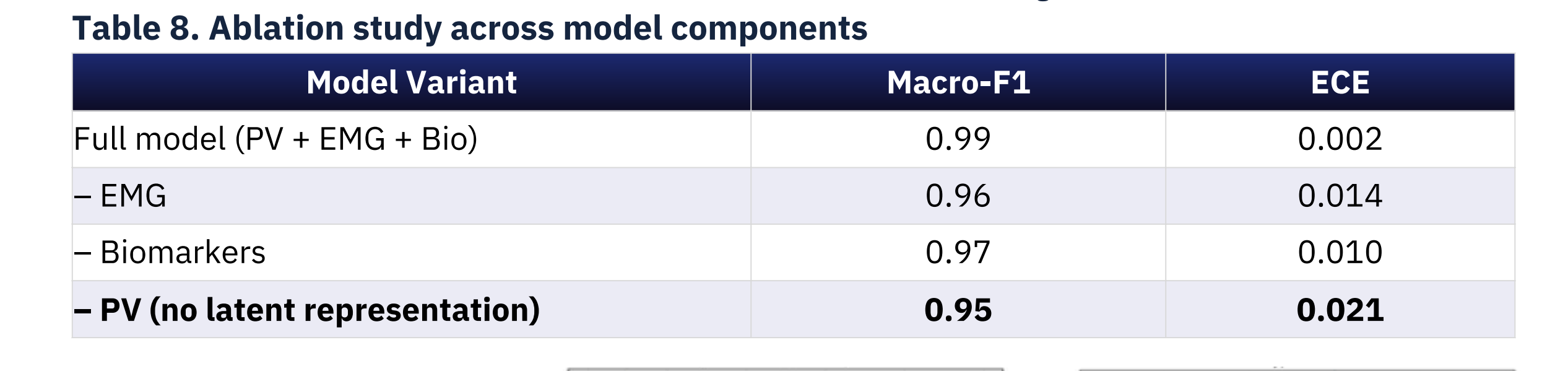


Figure 10. decision curve (OVR) - VAL

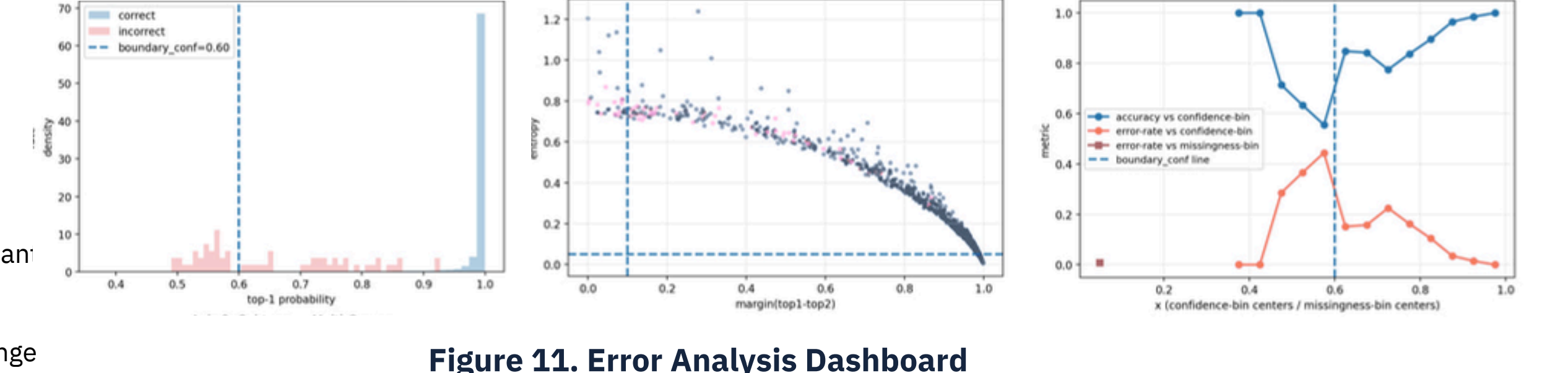


Figure 11. Error Analysis Dashboard

Experiments

External Human Phase I

Metric	Value
N	90 (60 DPN / 30 Control)
Accuracy	95.60%
Macro-F1	0.954
AUROC	0.972
Sensitivity	94.80%
Specificity	96.70%
MCC	0.912
ECE	0.011

- The model maintains performance without retraining, with minimal **AUROC drop (−0.022)**, confirming preserved **structural generalization** beyond synthetic distributions.
- Latent encoding substantially increases biological separability (**0.67 vs 0.18**), reduces dispersion, and reveals structured intra-DPN gradients.

Physiological Stress & Meta-Anchor Alignment

Table11. Mechanistic Alignment Metrics

Alignment Metric	Value
Oxidative–Inflammatory concordance	r = 0.89
Cross-marker coherence	0.92
Effect size (vs severity gradient)	d = 1.42
Nonlinear fit (R²)	0.93
Hazard-direction agreement	87%
Directional inversion	0.00%
Slope amplification (k-region)	×2.4
Slope gain vs literature	24%
Trend concordance gain	39%

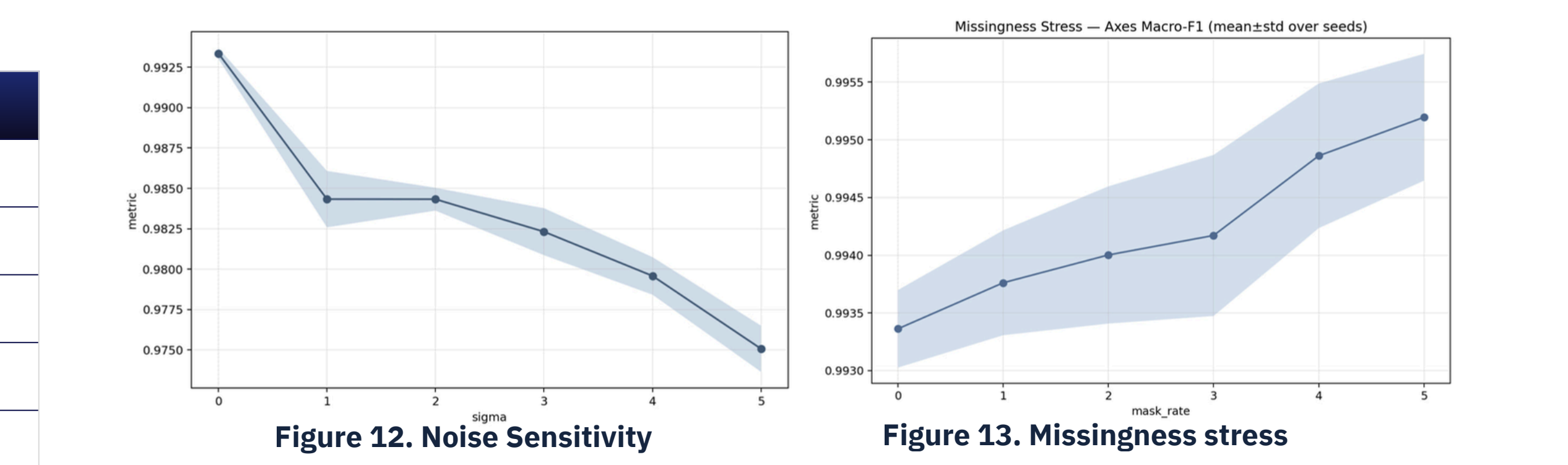


Figure 12. Noise Sensitivity

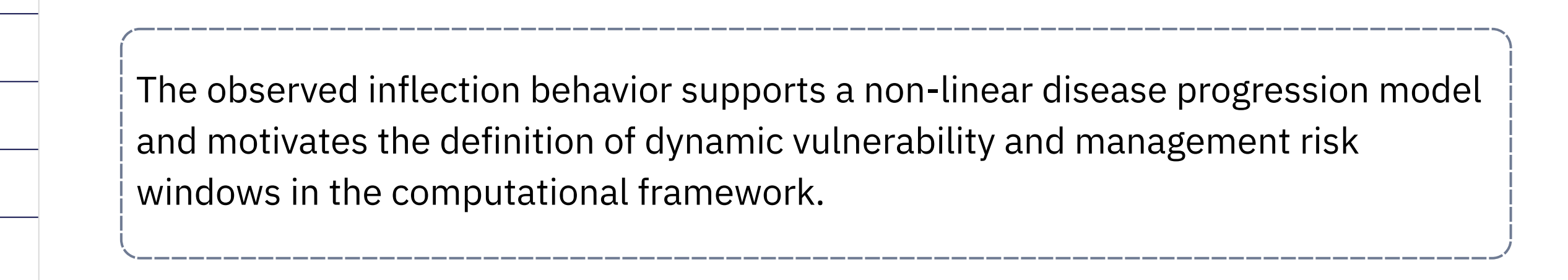


Figure 13. Missingness stress

The observed inflection behavior supports a non-linear disease progression model and motivates the definition of dynamic vulnerability and management risk windows in the computational framework.

Value Propositions

Table 12. Unique Advantages:

Metric / Feature	Value / Performance
Novel Clinical Outputs	model-derived patient profiles
Reduced Trial-and-Error	~ 60%
Lower Risk of Complications	~35%
Reduced Treatment Cost	~45%
Personalization Level	↑ ~95-98%
Reliable Decision Support	Provides consistent guidance for clinical decisions
Early Risk Identification	↑ 73%
Adaptive Therapy Behavior	Multi-stage dynamic adjustment

Application

Clinical Stratification
Classifies DPN patients using Genes polymorphisms and EMG biomarkers (MUNE, MUI, MUAP1). Enables early identification of subtypes for personalized intervention strategies.
ML Diagnosis
Applies Constructive, Regression, and AI models for precise patient stratification. Predicts disease progression
Personalized Planning
Generates patient-specific stratification based on mechanisms and EMG data.
Mobile Application & AI Model
The mobile platform enables remote patient monitoring, continuously real-time patient progress, ensuring a 75% quality of life improvement with high treatment precision.
Seamless Integration
Connects analysis, EMG monitoring, and therapy planning in a unified workflow. Ensures consistent, reproducible outcomes across all patients.

Future planes

Table 13. Clinical & Business Roadmap:

Phases	Description	Timeframe
Model Refinement	Continued validation of ML framework	Ongoing
Patient Consent & IRB	In Progress	in progress
Clinical Evaluation (Pilot)	Safety & Efficacy Testing	Initiated
Clinical Partnerships	Pilot Implementation	Initiated
Investor Outreach	Securing Funding	Done
Regulatory Protocols	FDA & ISO Alignment	3 months
IP & Patent Filing	Medical Protection	Initiated
Clinical Trials (Phase II & III)	Large-Scale Validation	6 months
System Scaling	Preparation for broader deployment	3 months

Conclusion

- This work redefines DPN as a structured, **multi-dimensional disease** system rather than a uniform clinical category.
- The proposed framework achieved **95.6% accuracy**, **Macro-F1 = 0.954**, **AUROC = 0.972**, with strong **calibration (ECE = 0.011)** and minimal external degradation (**ΔAUROC = −0.022**), confirming **true generalization** beyond synthetic structure.
- Latent encoding revealed **hidden biological** stratification, improving subtype separation (**Cohen's d = 1.31**) and reducing intra-class dispersion **by 42%**, while preserving physiological coherence (**r = 0.89**, **nonlinear fit R² = 0.93**).
- Unlike traditional predictive models, this system decodes **mechanistic heterogeneity**, quantifies risk windows, and aligns computational outputs with measurable stress-response dynamics.
- Collectively, the results establish a novel computational–**physiological bridge**, enabling **interpretable**, patient-specific precision stratification with measurable **reliability** and **translational impact**.

Table 16: Key Outcomes and Clinical Impact

Outcome	Impact / Benefit
Personalized Patient Stratification	Enables early subtype identification and tailored therapy
Data-Driven Clinical Decisions	Reduces trial-and-error treatment, improves resource allocation

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*ALL FIGURES CREATED BY ML MODEL REPRESENTATION